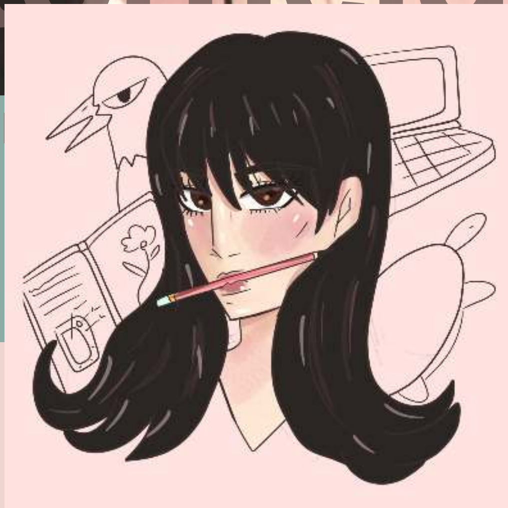

(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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The Human Transport System

The Circulatory System

▼ Explain why animals need a mass transport system:

- Animals have a **high metabolic requirement** and therefore needs essential molecules such as **glucose and oxygen** transported fast to cells during **respiration**.
- However, animals have a **small surface area to volume ratio** so **diffusion is not enough** to transport substances. To **overcome the limitations of diffusion**, there is a mass transport system where **fluid flows in one direction** in bulk **mass flow** to transport substances all over the body.

▼ Relate the structure of blood vessels to their functions:

Arteries

Arteries carry blood away from the heart at high pressure over long distances (except for pulmonary artery)

Structure - Function:

1. Narrow lumen - Maintains high blood pressure
2. Has endothelium lining - Protects the artery from damage. Provides a smooth surface so there is less resistance to blood flow.
3. Thick layer of smooth muscles - contracts and relaxes (vasoconstriction and vasodilation) to redistribute blood in the arteries. Helps maintain high blood pressure.
4. Thick layer of elastic fibres - stretches and recoils to maintain high pressure and rapid flow of blood
5. Thick collagen walls - Gives strength and helps withstand the high pressure of the blood

Veins

Veins carry blood into the heart at low pressure (except for pulmonary vein)

Structure - Function:

1. Wide lumen - Maintains low blood pressure
2. Has endothelium lining - Protects the vein from damage. Provides a smooth surface so there is less resistance to blood flow.
3. Thin layer of smooth muscles and elastic fibres - Since there is less pressure to resist and maintain
4. Collagen Fibres - Provides strength and support
5. Contains valves - prevents the back flow of blood at low pressure

Capillaries

Capillaries carry blood at low pressure to allow the exchange of substances between cell membranes

Structure - Function:

1. lumen - Maintains low blood pressure, prevents resistance
2. Has endothelium lining - keeps the short distance of diffusion
3. 1 cell thick wall - allows the diffusion of substances

The Heart Structure

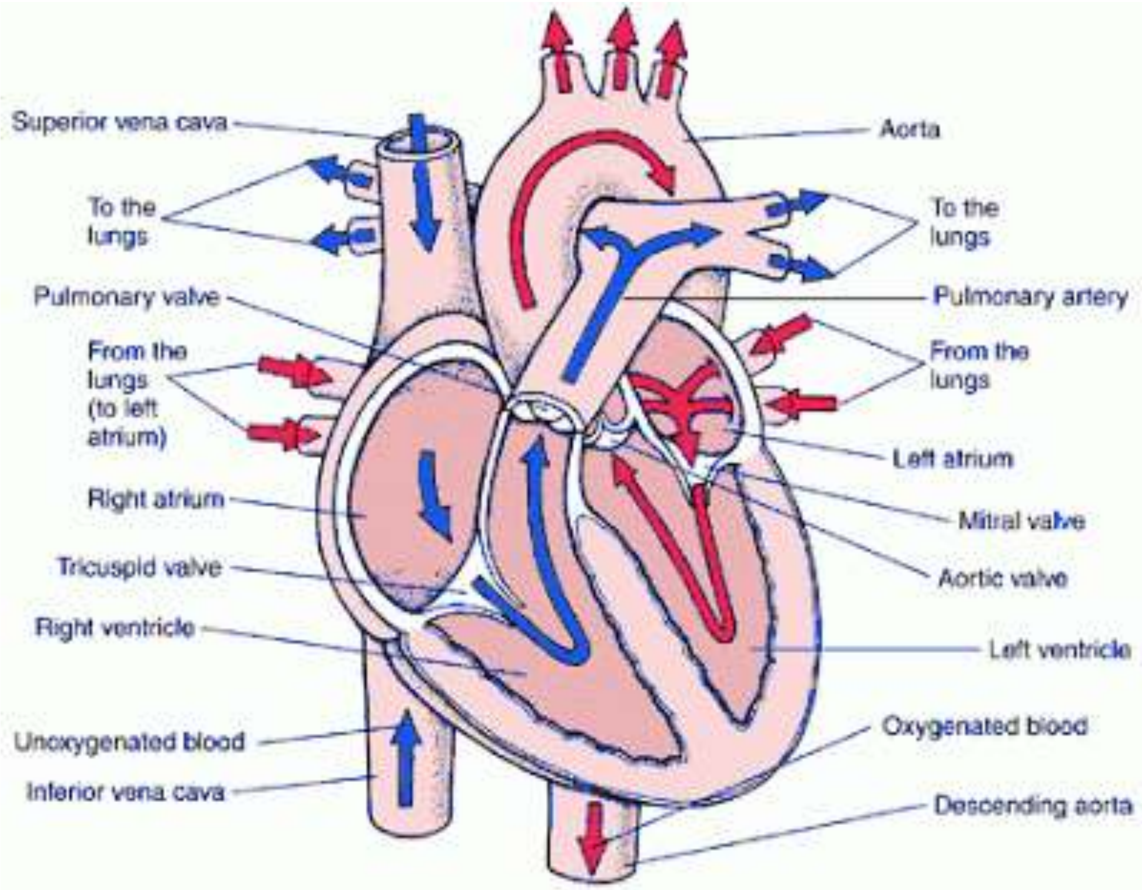
▼ Describe the structure of the heart:

- Made up of cardiac muscles/myocytes (muscle cells) that contract using myogenic contractions, meaning that the **heart's stimulation is made within the heart itself and leads to depolarisation.**
- It has **4 chambers**: the right and left atria and the right and left ventricles.
- The left and right side of the heart is separated by the septum. The septum separates deoxygenated and oxygenated blood.
- The walls of the left ventricle has **thicker, more muscular walls** than the rest of the heart because it needs to pump blood at the highest pressure over the longest distance to the body.
- Blood Vessels: Connecting to the left ventricle, there is the **aorta**, a major artery which carries blood from the left ventricle to the rest of

the body. Connecting to the right ventricle is the **Pulmonary Artery,** a major artery which takes blood to the lungs to be oxygenated. Connecting to the left atrium are **Pulmonary Veins** which bring oxygenated blood from the lungs back to the heart. And connected to the right atrium is the **Vena Cava - a** Major vein which brings deoxygenated blood from the body back to the heart. There are also **coronary arteries** on the outside of the heart which supplies the heart with oxygen.

- Valves: Between the atria and ventricles, there are the **atrioventricular valves —** bicuspid on the left and tricuspid on the right. When blood flows in from the pulmonary veins and vena cava into the atria, the atrioventricular valves open to allow the flow of blood into the ventricles. Between the left venticle and the aorta, there is the **aortic valve** that open to allow blood to be pumped into the aorta or close to prevent backflow of blood. Between the right ventricle and pulmonary artery there is the **pulmonary valve** which opens to push blood into the pulmonary artery, or close to prevent backflow. Pulmonary and aortic valve = Semi-lunar valves.

Question Number	Answer	Mark
*4(a)QWC	**Take into account quality of written communication when awarding the following points.** 1. idea that there are four chambers ; 2. correct reference to relative position of *atria* and *ventricles* ; 3. idea of left and right sides separate / *septum* ; 4. reference to muscular nature of walls ; 5. reference to *cardiac* muscle ; 6. idea of relative thickness of *ventricle* (walls) ; 7. correct reference to position of {*atrioventricular valves* / eq} ; 8. correct reference to position of *semilunar valves* ; 9. reference to position of {*tendons / tendinous cords / papillary muscles* / eq} ; 10. correct reference to position of {*aorta / pulmonary artery*} ; 11. correct reference to position of {*vena cava / pulmonary vein*} ; 12. correct reference to *coronary arteries* ; 13. reference to {*SAN / Sino Atrial Node / pacemaker/ AVN /Atrioventricular Node / Purkinje fibres /Purkyne fibres / Bundle of His/eq* } ;	**(5)**



▼ How is the heartbeat controlled? (Mention the SAN and the AAN)

The sinoatrial node (pacemaker) in the right atrium sends electrical impulses which causes the atria to contract. These electrical impulses are then sent to the atrioventricular node, where it passes through purkyne purkinje fibres to the ventricles, where they cause the ventricles to contract. Through this system, the sinoatrial node maintains heartbeat and ensures that the ventricles contract after the atria.

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/f9cc2365-cd16-4c84-aa2e-9bc5ca8d37b4/Structure__Function_of_the_Mammalian_Heart.pdf

The Cardiac Cycle

▼ Outline the Cardiac Cycle

Step 1: Atrial Systole & Ventricular diastole

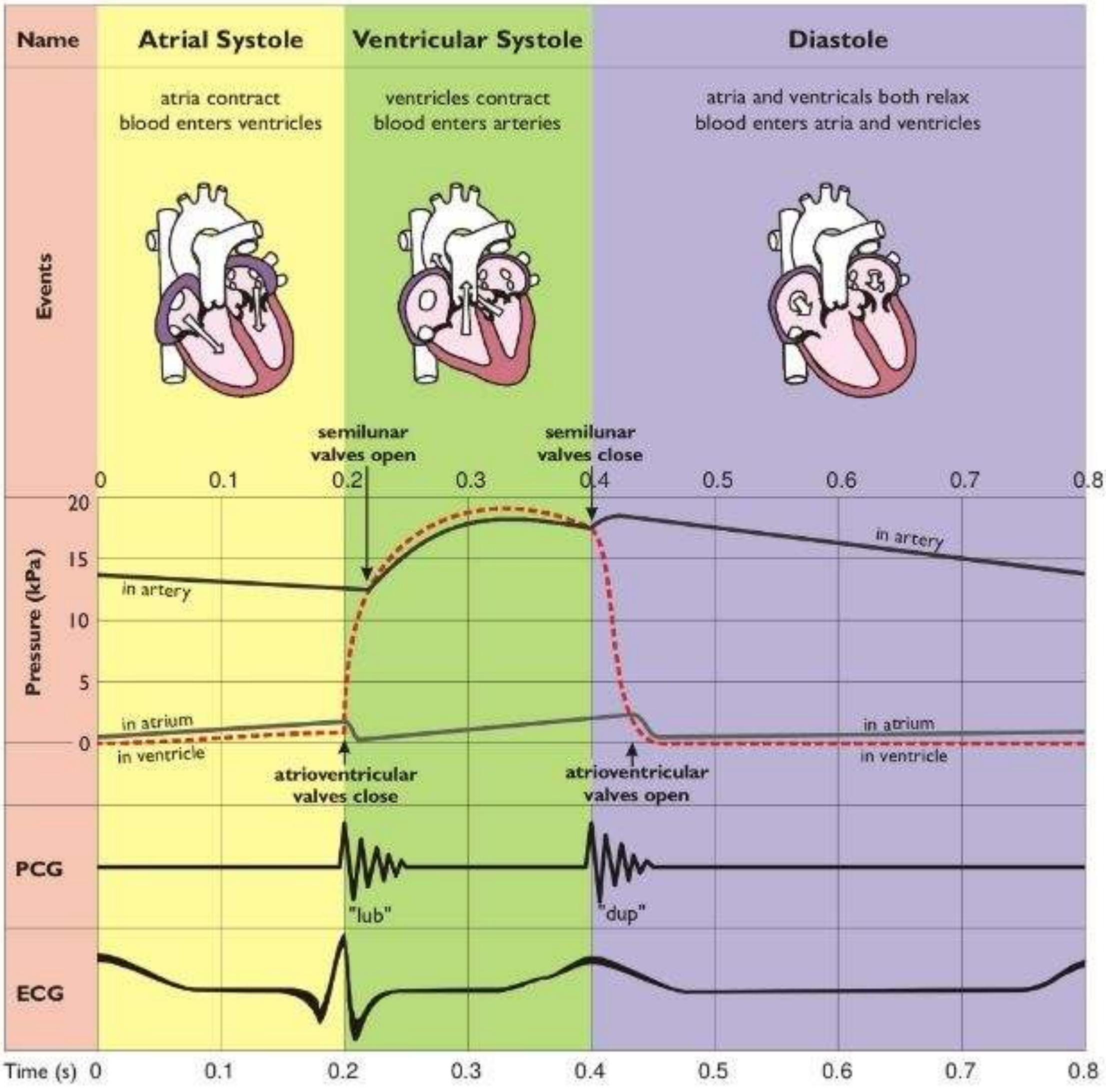
1. Blood enters the atria through the pulmonary vein and vena cava, the pressure and volume in the atria increase due to the entry of blood.
2. The atria contracts (atrials systole) and the increased pressure in the atria causes the atrioventricular valves to open and blood to be ejected into the relaxed ventricles. The semi-lunar valves close.
3. As blood is ejected from the atria into the ventricles, atrial volume and pressure begins to decrease.

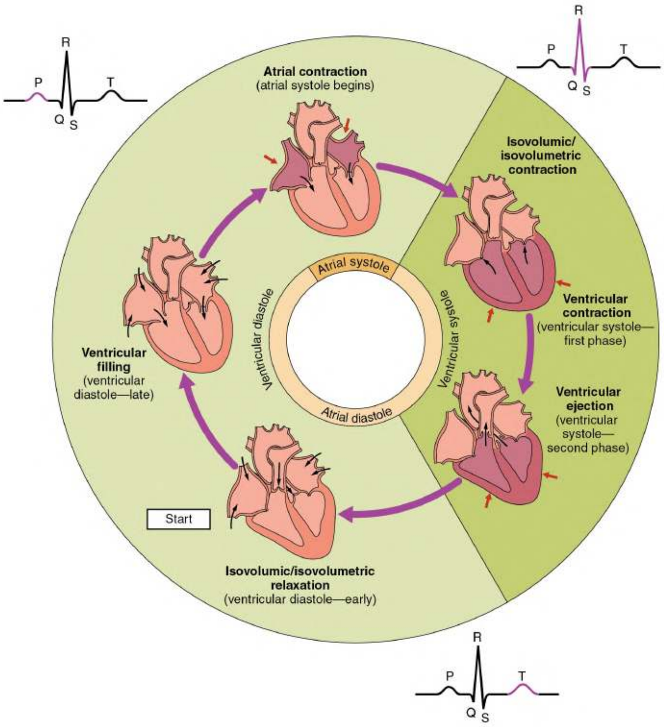
Step 2: Ventricular systole and atrial diastole

1. The ventricles fill up with blood entering from the atria through the atrioventricular valves. The ventricles contract causing pressure in the ventricles to increase. Blood is pushed out into the aorta and pulmonary arteries through the semi-lunar valves that now open. Volume in the ventricles begin to decrease
2. The atrioventricular valves close to prevent the back-flow of blood.

Step 3: Atrial and Ventricular Diastole/Cardiac Diastole

1. High aortic pressure causes **semi-lunar valves to close** to prevent back flow as blood leaves the heart.
2. When blood comes back to the atria via the pulmonary artery and the vena cava, the atria fills up with blood passively as aortic pressure is higher than atrial pressure.
3. Pressure in the atria increase, pressure exerted by the blood causes the **AV valves to open.**
4. Blood flows passively into the ventricles.


Name
Atrial Systole
Ventricular Systole
Diastole
Events
atria contract
blood enters ventricles
ventricles contract
blood enters arteries
atria and ventricals both relax
blood enters atria and ventricles
semilunar valves open
semilunar valves close
Pressure (kPa)
in artery
in atrium
in ventricle
atrioventricular valves close
atrioventricular valves open
PCG
"lub"
"dup"
ECG
Time (s) 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8



▼ Explain how an Electrocardiogram (ECG) Works

To Calculate a person's heart rate:

- An ECG measures the electrical activity of the heart.
- One heart beat is counted from one wave peak to the next of its kind i.e. from one QRS complex to the next QRS complex would be counted as one heart beat.
- To calculate the person's heart rate, you have to count the number of peaks in a given amount of time, then divide the number of beats by time (60 sec) to get a value for the heart rate.

Question Number	Answer	Additional Guidance	Mark
2(b)	1. idea that it shows electrical activity of the heart ; 2. idea of how to identify {one heart beat / time for one heart beat} ; 3. count the number of { these / peaks / eq } in a {set time / stated time} or how long from one set of electrical activity to the next ; 4. description of how to obtain heart rate e.g. beats divided by time ;	**ACCEPT for 2:** from one {P wave / QRS complex / T wave } to the next	(3)

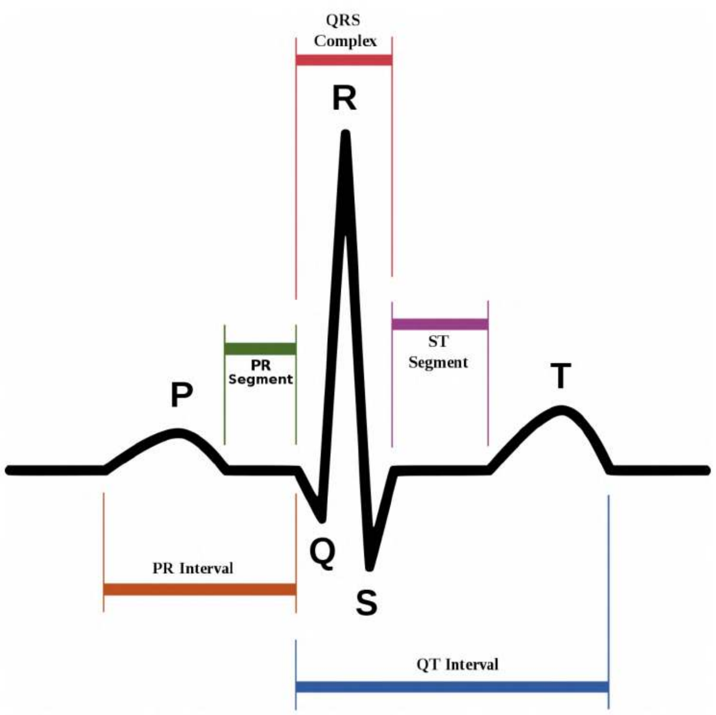


Figure 5: The start is diastole where both the atria and ventricles are relaxed. The P wave shows the depolarisation of the atria and is followed by atrial systole (contraction). Atrial systole continues to the end of the QRS complex and the point where the atria relax. The QRS complex shows the relaxation of the ventricles and is followed by ventricular systole (contraction). The T wave represents the a further depolarization (re- polarization) and marks the start of ventricular relaxation.

Figure 5: The start is diastole where both the atria and ventricles are relaxed. The P wave shows the depolarisation of the atria and is followed by atrial systole (contraction). Atrial systole continues to the end of the QRS complex and the point where the atria relax. The QRS complex shows the relaxation of the ventricles and is followed by ventricular systole (contraction). The T wave represents the a further depolarization (re- polarization) and marks the start of ventricular relaxation.

A normal healthy ECG trace shows what is known as the “PQRST shape” (Figure 5). Wave P is the stage that corresponds to atrial systole, excitation and contraction of the atria. Ventricular systole forms the “QRS” complex and the start of ventricular diastole corresponds to the “T” wave.

Oxygen Transport in Blood

▼ What is the role of Haemoglobin in oxygen and carbon dioxide transport?

Structure of Haemoglobin is a

- Water-soluble globular protein found in erythrocytes. Has 4 subunits - 2 beta polypeptide chains and 2 alpha polypeptide chains, each bound to a haem group containing Fe+ atoms (4 haem groups in total).
- Haem groups have high oxygen affinity; they bind to oxygen to form oxyhaemoglobin. Thus, 4 molecules of oxygen in total can bind to one haemoglobin.
- The haem group in haemoglobin binds/loads (high affinity) to oxygen to form oxyhaemoglobin at high partial pressure (high oxygen

concentration) and unbinds/unloads (low affinity) with oxygen at low partial pressure to release oxygen in respiring cells.

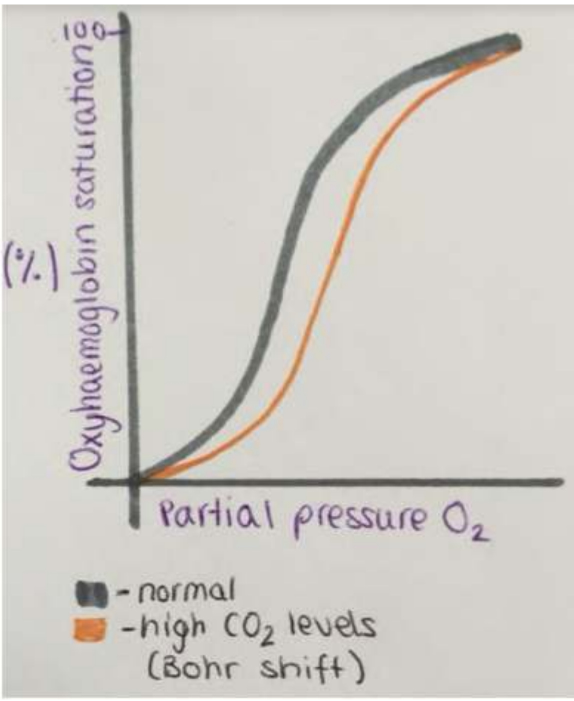
- At the same time, in low partial pressure, carbon dioxide binds to haemoglobin to form carboxyhaemoglobin. The deoxygenated blood returns to the lungs where carbon dioxide unloads from haemoglobin, which binds to oxygen again.

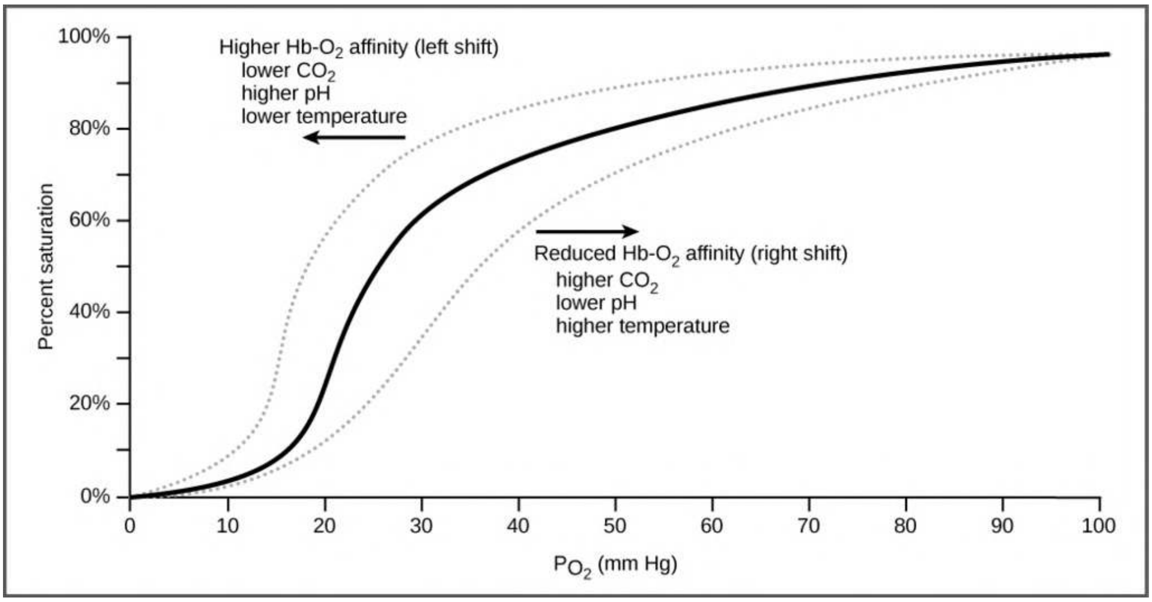
Oxygen release from oxyhaemoglobin depends on the demand for oxygen of tissues.

▼ Describe the Oxygen Dissociation Curve and the Bohr Effect

The Oxygen Dissociation curve illustrates the change in haemoglobin saturation as partial pressure changes.

- When partial pressure is high, haemoglobin has a high oxygen affinity and therefore is saturated. Conversely, when partial pressure is low, haemoglobin has lower oxygen affinity as the demand for oxygen causes oxygen to dissociate from oxyhaemoglobin. This causes it to be lowly saturated.





- The "S" Shape of the oxygen dissociation curve is due to the co-operative binding nature of haemoglobin and oxygen. Where the binding of the first oxygen molecule is harder but it alters the shape of the haemoglobin molecule, making it easier for the following oxygen molecules to bind to it. In the same way, when one oxygen molecule dissociates, the following 2nd, 3rd and 4th dissociate more readily.

The Bohr effect - The Bohr effect - Haemoglobin has higher affinity to carbon dioxide than oxygen. Thus, at high partial pressure of carbon dioxide, haemoglobin has a lower affinity for oxygen even at high partial pressure of oxygen, causing it to be released into the respiring cell that needs it, and consequently, carbon dioxide to be taken in and bound to haemoglobin and carried away to the lungs.

Note* Increase in CO2 conc. = Lower pH

▼ Differentiate between Fetal and Adult Haemoglobin

By the time oxygen reaches the placenta, the partial pressure of the placenta has already decreased. Thus, to survive, fetal haemoglobin needs to have a higher oxygen affinity than adults, as it needs to be able to absorb oxygen at low partial pressure for respiration.

- **Fetal haemohlobin has higher oxygen affinity and can bind to oxygen at lower partial pressure** than adult haemoglobin because it needs to be able to absorb oxygen at low partial presssure for respiration, since by the time oxygen reaches the placenta, the partial pressure has already decreased.

Cardiovascular Disease

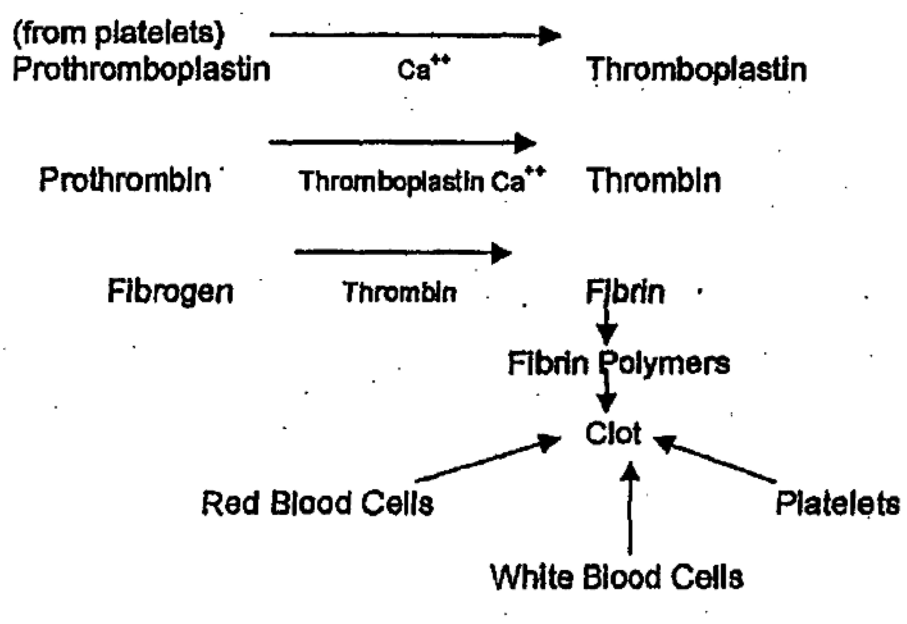
▼ Describe the sequence of events that lead to atheroscelerosis

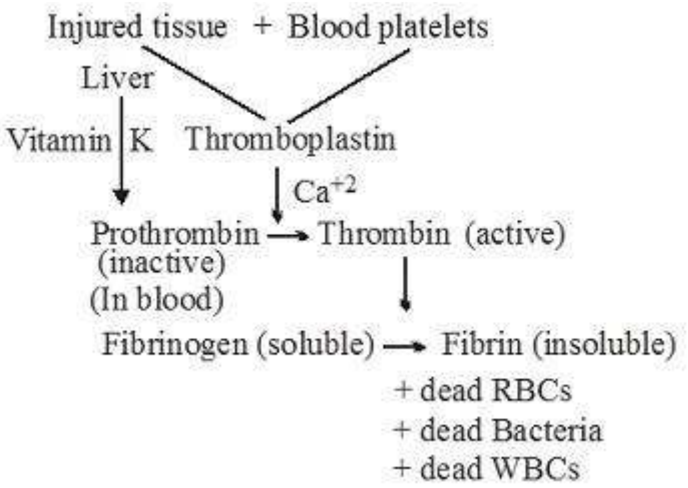
Atherosclerosis is the hardening of arteries caused by the build-up of fibrous plaque called an atheroma which leads to further CVD :

1. Endothelial Dysfunction: The endothelial lining of the arteries is damaged due to factors such as high cholesterol levels, smoking or high blood pressure.
2. Inflammatory Response: Damage to the endothelium lining increases the risk of blood clotting and leads to an inflammatory response which causes WBCs to move into the arteries.
3. Plaque Formation: Over time, the WBCs, cholesterol, calcium salts and fibres build up and harden, leading to plaque formation.
4. Raised Blood Pressure: Plaque formation causes the arteries to narrow, lose their elasticity, and restricts blood flow, and thus increases blood pressure. The increased BP then damages the endothelial lining causing the process to be repeated.

▼ Describe the formation of blood clots

When vessels are damaged, blood clots form to prevent blood loss and the entering of pathogens into the bloodstream that could cause infection. HOWEVER, blood clots can also damage the body by forming inside blood vessels and restricting blood flow, leading to thrombosis.





Blood clotting:

1. Platelets attach to the exposed collagen fibres when they come in contact with the damaged blood vessel.
2. Platelets then release a protein known as Thromboplastin which converts inactive prothrombin into active thrombin in the presence of calcium ions and Vitamin K.
3. Prothrombin catalyses the conversion of soluble fibrinogen into insoluble fibrin.
4. Fibrin forms a mesh of fibres which trap the red blood cells, and thus form a clot.

▼ Factors Causing CVD

1. **Genetics** - certain genes can increase the risk, sometimes indirectly for instance by having genes for a higher blood pressure. **Family history** of the disease also increases your risk.
2. **Obesity -** When energy intake is more than energy usage of the body, the excess energy is converted to fats in the liver and stored in fat stores in the body as LDLs. This increases mass in the body leading to an increased level of pressure put on the cardiovascular system as the heart needs to work harder to pump blood to an increased number of cells. This leads to high blood pressure which can lead to damage to the endothelial lining and atheroscelerosis.
3. **Diet** - diets high in **cholesterol (saturated fats)** increase the build-up of **plaque** on arteries. **High sodium/salt diets** cause an increased blood pressure as the high solute/salt concentration causes more water to be

drawn up into the bloodsteam, which increases the volume of blood and thus pressure.

4. **Age** - prevalence of CVD increases with age as blood vessels lose their elasticity over time.
5. **High blood pressure** - this can **narrow** and damage arteries by causing atheroscelerosis or cause an **aneurysm**, both of which increase the risk of CVD
6. **Smoking** - smoking **damages the lining** of arteries and can cause the formation of
atheromas.
7. **Inactivity** - has been linked with an increase in blood pressure
8. **Gender -** the hormone estrogen reduces the risk for CVD in women compared to men. In addition, due to their hormones, men are more likely to store fat in their abdomen which is linked to increased chances of CVD.

Thus risk of CVD can be reduced by stopping smoking, regular exercise, reducing consumption of

alcohol, dietary changes and maintaining healthy body weight.

▼ Outline the link between dietary antioxidants and CVD

Oxidative stress is an imbalance of **antioxidants** and **free radicals** of oxygen in the body, which are oxygen atoms with an **uneven number of electrons**, making it **highly reactive** and meaning it can cause **damaging chains of chemical reactions** in the body. This is believed to cause CVD by causing damage to the artery walls. Antioxidants can stabilise the oxygen radical by **donating electrons.** Thus, the intake of additional antioxidants in the diet should help **prevent** some cases of Cardiovascular disease and at least lessen the risk.

▼ Describe the significance of lipoproteins in maintaining blood cholesterol levels

Lipoproteins found in fatty foods or formed in the liver are used to transport cholesterol and maintain blood-cholesterol levels. There are 2 types: **high-density lipoproteins (HDLs)** and **low-density lipoproteins (LDLs)**. HDLs transport cholesterol to the liver whereas LDLs transport them to blood vessels.

High-density lipoproteins

Formed from unsaturated fats and proteins, reduces cholesterol levels in the blood by transporting cholesterol to the liver to be **expelled.**

Low-density lipoproteins

Formed from saturated fats and proteins, increases cholesterol levels in the blood by transporting cholesterol to the arteries where it can build up and form plaques, leading to atheroscelerosis.

HDL > LDL = 😊

There is a **positive correlation** between saturated fat intake and the increase in cholesterol
levels. If SF ⬆then CL ⬆, which means → LDLs cause cardiovascular disease.

▼ Obesity indicators

Obesity indicators such as BMI and Waist to hip ratios are used to evaluate the effect of diets in causing CVD and can help encourage dietary changes.

Body mass index (BMI)

BMI is a value calculated using the following equation:

Body mass in Kilograms ÷ (body height in metres)2

The value generated can be compared to a chart which classifies you under the following:

- Under 18.5 - underweight
- 18.5 - 25 - normal
- 25 - 30 - overweight
- Over 30 - obese

  0.9=< obese for males, 0.85< obese for females.

Waist to hip ratio (WHR)

Calculated by dividing the waist measurement to hip measurement. Women ideally should have a WHR below 0.85, whereas men should have it below 0.9.

▼ Why do people overestimate/underestimate risk?

Risk is the chance of something unfavourable occurring. Statistical chance of something occurring can be supported using data obtained by scientific research. Whereas a **perceived risk** varies from person to person and is based on **factors,** such as approval of activity. Therefore, as a result of that, the perceived risk can **vary greatly from the actual risk,** thus leading to **underestimating or overestimating** the probability of occurrence of an unwanted event or outcome.

▼ Outline the treatments available for CVD

1. **Antihypertensives -** drugs that **lower blood pressure.** Inexpensive **but** may have minor side effects - Diuretics that increase volume of urine and beta blockers that block the heart's response to hormones i.e. adrenaline.
2. **Statins - Drug** used to **lower cholesterol levels** by inhibiting cholesterol production. Increases HDL : LDL ratio.

   PRO: Mostly **effective,** helps lower BP by allowing blood vessels to relax → helps prevent CVD

   CON - can cause **nausea, vomiting and aches in muscles and joints**, and sometimes **diabetes.**
3. **Plants stanols and sterols -** Reduces the amount of cholesterol absorbed from the gut to the blood. Reduces amount of LDLs in the blood.
4. **Anticoagulants -** drugs that slow down the body's process in forming clots by inhibiting the synthesis of clotting factors (i.e. prothrombin) and thus **prevent blood clotting.**

   PRO: Prevents thrombosis which reduces blood flow in the artery, arising from formation of internal clots

   CON - if blood vessels are damaged and blood clotting takes a long time to occur, excessive internal bleeding can cause a haemorrhage
5. **Platelet inhibitors -** drugs that prevent blood clots forming by making platelets less sticky and thus able to stick to exposed collagen of damaged vessels or clump to eachother to form clots.

   PRO - prevents the formation of blood clots in arteries that anticoagulants are ineffective at preventing.

CON - Can lead to excessive internal bleeding and **haemorrhage** due to **slow** clot formation.

▼ Differentiate between correlation and causation

- Quantitative data - Data that can be represented with **figures**. For instance, your weight.
- Correlation - correlation describes **the relationship between 2 variables**. A value of 1
  means a perfect positive correlation (as one increases, the other also increases). A value of 0 is no correlation and a value of -1 is perfect negative correlation (as one increases, the other equally decreases). For instance, there is a positive correlation between cholesterol levels and the number of cases of CVD.
- Causation - This is when **one variable causes another,** and is different to correlation. 2 values can have a correlation, but that does not mean they cause each other. For instance, an increase in cholesterol levels can cause an increase in plaque formation (causation and correlation); or correlation only - the average temperature in 2 places both increase over 2 months, this is correlation, but one has not caused the increase in the other, so there is no causation.

▼ How do you evaluate studies?

- In order to make conclusions from scientific studies they must have been carried out appropriately to **avoid bias** and to be **representative** of the whole population.
- Sample selection must be done **randomly** to avoid bias, it must be a **large** enough sample size to be representative of the whole population and must also be sampled across different areas. For instance a nationwide study couldn’t sample from one city only.
- A reliable study has **statistical analysis** and **peer review** from other scientists. Any trials involving patients and doctors should have a **control group, a placebo, and be blind - or ideally double-blind**.
  - Statistical Analysis: Using data to create Comparisons and figure out trends.
  - Peer Review: Process by which studies are reviewed by a separate expert to ensure there are no errors or bias in the study

- Control Condition: Condition that lacks any manipulation of an independent variable or is not exposed to the IV, which is used to ensure that any the experiment is valid and that any changes in the experimental condition is due to the IV.
- Placebo: An inactive substance that resembles a drug but has no effect (on the body)
- Double-blind - Technique where the subject and the experimenter don't know which condition they are put under, which allows an unbiased evaluation of results.

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